ECE 310 Fall 2026

Lecture 2: Math preliminaries, complex numbers

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Learning Objectives

- Explain what defines a complex number, be able to identify the magnitude and phase of a complex number, and visualize them on the complex plane.
- Convert between rectangular and polar forms of complex numbers.
- State Euler's identities for complex numbers.
- Interpret and visualize common discrete-time signals including Kronecker delta and unit step.



Complex Numbers: Rectangular Form

For a complex number $x \in \mathbb{C}$, we may express x in *rectangular form*

$$x = a + jb$$

- a: real part
- b: imaginary part
- j: imaginary unit

$$j = \sqrt{-1} \quad , \quad j^2 = -1$$

We may also denote x^* as the *complex conjugate* of x where $x^* = a - jb$.

Complex Numbers: Polar Form

For a complex number $x \in \mathbb{C}$, we may express x in *polar form* = exponential form

$$x = Re^{j\theta}$$

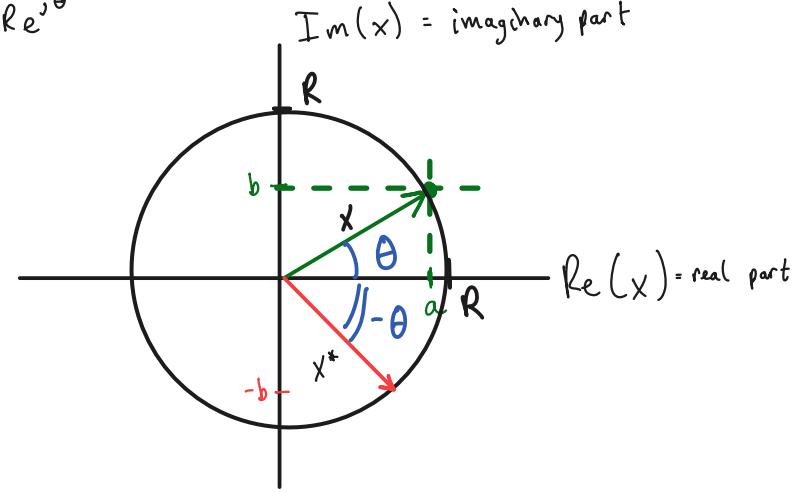
- R: length
- θ: angle
- j: imaginary unit

We may also denote x^* as the *complex conjugate* of x where $x^* = Re^{-j\theta}$.



Visualizing Complex Numbers

Let $x = a + jb = Re^{j\theta}$



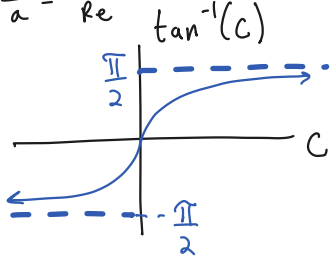
Magnitude and Phase

Complex numbers are described by two important quantities known as *magnitude* and *phase*. Let $x = a + jb = Re^{j\theta}$

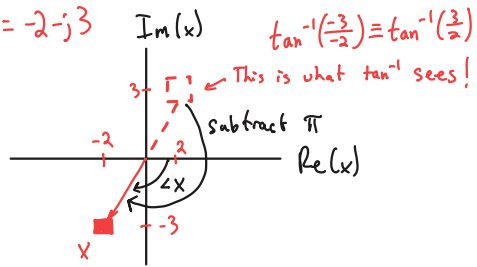
- $\text{Magnitude}(x) = |x| = \left|\sqrt{a^2 + b^2}\right| = \left|\sqrt{xx^*}\right| = R \geq 0$
- $\text{Phase}(x) = \angle x = \begin{cases} \tan^{-1}\left(\frac{b}{a}\right), & a \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) + \pi, & a < 0,\ b \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) - \pi, & a < 0,\ b < 0 \end{cases} = \theta \in [-\pi, \pi]$.

(Quadrants 1 and 4) (Quadrant 2) (Quadrant 3)

$C = \frac{b}{a} = \frac{Im}{Re}$



Ex: $x = -2 - j3$



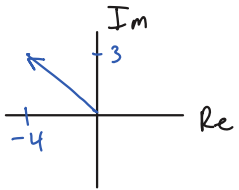
Magnitude and Phase Examples

Determine the magnitude and phase of the following complex numbers and sketch them on the complex plane.

(a) $x = -4 + j3$

(b) $y = j2$

(a)



$|x| = \sqrt{(-4)^2 + (3)^2}$

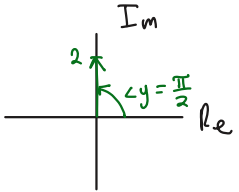
$= \sqrt{25}$

$= 5$

$\angle x = \tan^{-1}\left(\frac{3}{-4}\right)$

$= \tan^{-1}\left(-\frac{3}{4}\right) + \pi$

(b)



$|y| = \sqrt{(0)^2 + (2)^2}$

$= 2$

Magnitude and Phase

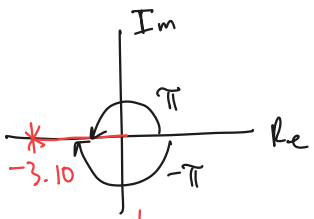
Complex numbers are described by two important quantities known as *magnitude* and *phase*. Let $x = a + jb = Re^{j\theta}$

- $\text{Magnitude}(x) = |x| = \sqrt{a^2 + b^2} = \sqrt{xx^*} = R \geq 0$
- $\text{Phase}(x) = \angle x = \begin{cases} \tan^{-1}\left(\frac{b}{a}\right), & a \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) + \pi, & a < 0,\ b \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) - \pi, & a < 0,\ b < 0 \end{cases} = \theta \in [-\pi, \pi].$

Special case to consider: What are the magnitude and phase of $x = -3.10$?

$x = -3.10 + j0$

$= 3.10\, e^{\pm j\pi}$



$a = -3.10$

$b = 0$

$|x| = 3.10$

*Real numbers can only have zero phase or ± π phase (if negative)

Complex Arithmetic: Addition and Subtraction

Let $x = a + jb = Re^{j\theta}$ and $y = c + jd = Se^{j\phi}$

$$x + y = (a+c) + j(b+d) = Re^{j\theta} + Se^{j\phi}$$

$$x - y = (a-c) + j(b-d)$$

Complex Arithmetic: Multiplication and Divison

Let $x = a + jb = Re^{j\theta}$ and $y = c + jd = Se^{j\phi}$

$$xy = (a+jb)(c+jd)$$

$$= ac + j^2 bd + jbc + jad$$

$$= (ac - bd) + j(bc + ad)$$

$$= Re^{j\theta} Se^{j\varphi} = RSe^{j(\theta+\varphi)}$$

$$\frac{x}{y} = \frac{a+jb}{c+jd} \cdot \frac{c-jd}{c-jd}$$

$$= \frac{ac + bd + j(bc - ad)}{c^2 + d^2}$$

real part

imaginary part

$$= \frac{Re^{j\theta}}{Se^{j\varphi}} = \frac{R}{S} e^{j(\theta-\varphi)}$$

Euler's Identities

Euler's formula allows us to convert from polar to rectangular forms for complex numbers. Let $x = Re^{j\theta}$.

$$x = R\cos\theta + Rj\sin\theta = Re^{j\theta} = a + jb$$

This leads to a few important identities:

- $e^{\pm j\pi} = -1$
- $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$
- $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$

$a = R\cos\theta$

$b = R\sin\theta$

Complex Arithmetic Examples

Simplify the following expressions into rectangular and polar forms.

(a) $j2 + \sqrt{2}e^{-j\frac{\pi}{4}}$

(b) $\frac{j3}{1-j}$



Let's consider an example length–7 signal:

$$x[n] = \{1, 4, -2, 0, 3, 1, 2\}$$

Is there something missing here?



Remember $n \in \mathbb{Z}$ (integers)

Let's consider an example length–7 signal:

$n \in \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$

$$x[n] = \{1, 4, -2, 0, 3, 1, 2\}$$

Is there something missing here? We don't know where the signal stops and ends!

$$x[n] = \{1, \mathbf{4}, -2, 0, 3, 1, 2\}$$

$$x[n] = \{1, \underset{\uparrow}{4}, -2, 0, 3, 1, 2\}$$

this is where $n=0$

$$x[n] = \{1, \underline{4}, -2, 0, 3, 1, 2\}$$

$$\{x[n]\}_{n=-1}^{5} = \{1, 4, -2, 0, 3, 1, 2\}$$

start @ $n=-1$ end @ $n=5$

$x[0] = 4$

$x[-1] = 1$

$x[1] = -2$

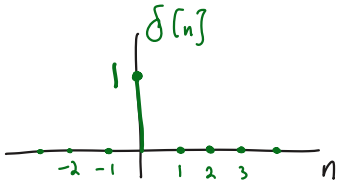
$x[2] = 0$

Kronecker Delta

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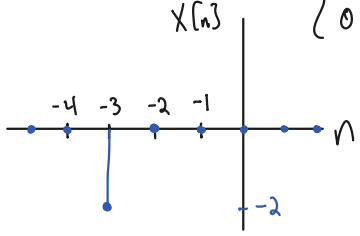
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$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} .$$



- Also known as unit impulse or unit pulse
- **This is not the same as the Dirac delta!**

Sketch $x[n] = -2\delta[n+3]$ = $\begin{cases} -2, & n=-3 \\ 0, & n \neq -3 \end{cases}$



Write $x[n] = \{2, 0, 3, -1, -4\}$ in terms of $\delta[n]$.

n=-2, n=-1, n=0, n=1, n=5

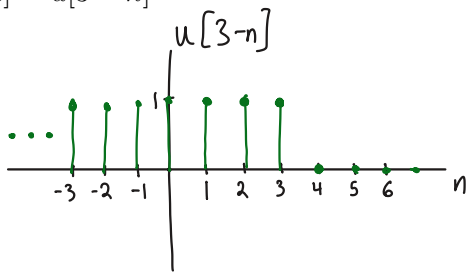
$$x[n] = 2\delta[n+2] + 0\delta[n+1] + 3\delta[n] - 1\delta[n-1] - 4\delta[n-2]$$

Unit Step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} .$$

- We can flip and shift just like with Kronecker deltas.

Sketch $x[n] = u[3 - n]$



$x[3] = u[3-3] = u[0] = 1$

$x[4] = u[3-4] = u[-1] = 0$

$x[2] = u[3-2] = u[1] = 1$

Sinusoids and Exponentials

- Sinusoids

$$x[n] = A\sin(\omega_0 n + \theta),\ A \in \mathbb{R}, \omega_0 \in \mathbb{R}, \theta \in \mathbb{R}$$

- Exponentials

$$x[n] = Ba^n, a \in \mathbb{C},\ B \in \mathbb{C}$$

